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Chapter 33: When the City's Clock Is Replaced—Real Estate, Finance, and the Discrete Leviathan

Abstract

This chapter characterizes the city as an "ecological clock"—a functional ensemble that provides cognitive calibration through bodily presence, everyday interaction, and multi-source observation—and distinguishes two de-anchorings: the financialization of real estate converts property into tradable readings, while the attention economy lets platform distribution and dissemination metrics participate in organizing the city's design, investment, and spatial presentation. The chapter applies the pseudo-small fiber mechanism defined in Chapter 4 to the urban scale—a state in which the certainty presented by platform readings exceeds the certainty they actually support—and characterizes its formation through a three-step manufacture: lowering operational friction, encoding urban space into standardized readings, and narrowing the contrast conditions needed for auditing. On the evidential level, the sectoral and geographic divergence of office demand, the association between the decline of independent shops and rising support for right-wing populism in the UK, and the spatial distribution of dark kitchens and dark grocery stores jointly indicate that urban functions are being partially reorganized; while the existing evidence is insufficient to prove that ecological functions have been completely replaced, the risk of irreversibility in long-term change warrants attention. The core of repair is to preserve and rebuild the spatial conditions that support independent observation, public interaction, and external verification.

Keywords

ecological clock; real estate financialization; attention economy; pseudo-small fiber; urban publicness; cognitive calibration

§1 The City Is an Ecological Clock

Within the framework of the Clock Room, the city is not a container—it is a kind of clock. But it is a special kind of clock: it simultaneously points to the sun.

The city processes information. The density of pedestrians on the street reveals whether this area is prosperous. The lights of the shops reflect whether the economy is active. The faces of neighbors show whether the community is stable. The number of children reflects the demographic structure. These signals are not indicators compiled by some institution—they are the direct outflow of the true state (T); the mapping function (π) is minimal, the fiber is extremely small, and they can be received by the body almost without decoding. When the body walks on the street, hunger, fatigue, security, and belonging are in direct contact with the person. This is existence in the time dimension—continuous, ungraduated, in direct causal contact with the real.

The city is also simultaneously a clock—it has graduations. The grid of the blocks, the routes of the commute, the hours of operation, the cycles of leases—these discretize continuous life. But unlike a purely financial clock, the fiber between the readings (r) of the city clock and the sun is small. Rent has gone up—the person directly feels that the purchasing power of wages has gone down (the mapping function is roughly known, the signal source is transparent). The shop on the corner has closed—residents can verify the cause of closure by asking around, long-term observation, and multiple sources of information (signal source transparent). The city's clock is anchored in physical reality—because the city is itself part of physical reality.

Lefebvre (1992) distinguished two kinds of time: *cyclical time*—the alternation of day and night, hunger, fatigue—belonging to the real time of the cosmos and of biology; and *linear time*—mechanical, quantifiable, clock time set for the pursuit of profit. The city is the intersection of the two times: the punch clock and the commuting timetable belong to linear time; the casual chat on the street corner and the gaze of the neighborhood belong to cyclical time. In a well-functioning city, both times coexist—the clock dimension and the time dimension are superimposed in the same physical space.

Mumford (1934) pointed out that the key machine of the modern industrial age was not the steam engine—it was the clock. The monastery bell tower was the first physical entity to rigorously divide human activity and rest into discrete units. But the monastery bell tower divided day and night according to fixed canonical hour rules (the sevenfold division of the Benedictine daily office), and this division still took sunrise and sunset as its reference, not yet having broken away from the natural framework delimited by the sun's movement. The mapping function (π) was transparent and stable—the person knew what the morning prayer bell signified, knew why it rang at that moment. The fiber was small.

The core function of the city as an ecological clock is this: it allows the person to operate within the clock dimension while preserving the channel for retreating back into the time dimension—the window is always open. The person works in an office building (clock dimension), goes downstairs and walks into the street (time dimension). Looks at data reports on the computer (readings), looks up and sees the weather outside the window (the sun). The physical space of the city provides the structural conditions for switching back and forth between

the clock dimension and the time dimension—the physical infrastructure of *flexibility*.

§2 The First De-Anchoring: The Financialization of Real Estate

Real estate is the heaviest physical entity in human society. Constructing a building requires a long construction period—steel and concrete have incompressible physical cycles. Purchasing a home requires thirty years of labor time—the monthly mortgage payment is anchored to wages and living costs. The value of real estate is embedded in the surrounding social network—school district, commuting distance, neighborly relations. These are all existences in the time dimension—full of heavy, slow, unacceleratable friction. Chapter 11 of the *Tao Te Ching* uses the bowl as an analogy: the walls of the bowl are "being"; the cavity enclosed by the walls is "non-being"—"it is exactly where there is nothing of it that the functionality of the vessel resides" (Lynn, 1999). Real estate is the physical embodiment of this "non-being"—a thirty-year holding period is precisely a cavity, within which information (life experience) is slowly deposited.

The financialization of real estate is the discretization of this heavy time into tradable clock readings (r).

First Layer of Financialization: Mortgage Loans. Mapping thirty years of labor time into a single monthly payment figure. The mapping function (π) is: (future thirty-year wage stream $\to$ one monthly repayment amount). This mapping is a dimensionality reduction—it discards everything the person might experience in those thirty years: unemployment, illness, divorce, industry decline. But the fiber of this first layer of mapping is relatively controllable—the bank, when assessing repayment capacity, at least examines proof of income, job stability, and credit history. The mapping function is lossy, but roughly known.

Second Layer of Financialization: Securitization. Mortgage loans are pooled into mortgage-backed securities (MBS); selected MBS tranches can then be repooled into collateralized debt obligations (CDOs), whose tranches may in turn be repackaged into CDO^2. Each layer superimposes a new mapping function. By the level of CDO^2, no single person can penetrate all the mapping layers and return to the sun—the simple fact of whether that borrower can repay the loan. The clock no longer points to the sun but to another clock—the credit rating of the tranches one layer up. Mapping self-reference is fully instantiated here.

Khandani, Lo, and Merton (2009) precisely described the mechanism by which financialization eliminates friction: the convergence of three trends in the U.S. residential real estate market—rising house prices, falling interest rates, and near-frictionless refinancing opportunities—led to a dramatic increase in systemic risk within the financial system. "Near-frictionless refinancing"—this is the precise financial counterpart of the "bowl" analogy above. Friction is the cavity of the bowl, "non-being." Eliminating friction is equivalent to filling the bowl solid. The bowl is still there, but it can no longer hold anything.

The BIS literature on financial cycles provides a precise temporal calibration for this de-anchoring process. Drehmann, Borio, and Tsatsaronis (2012) identified a distinct financial cycle—best characterized by the co-movement of medium-term cycles in credit and real estate prices, averaging around 16 years in duration, with cycles peaking after 1998 approaching 20 years. This 16–20 year medium-term fluctuation sits, in time scale, between the business cycle and generational change. On this basis, this chapter discusses the mismatch between financing rhythms and the city's long-term change—this is a description of rhythm mismatch, not a claim that financialization has "compressed" the generational rhythm.

Borio (2012) provided the leading indicators of crises: the most promising leading indicators of financial crises

are based on the simultaneous positive deviation of the credit-to-GDP ratio and asset prices (especially real estate prices) from historical norms. Translated into the language of the Clock Room: when the readings of the financial clock deviate sufficiently far from the position of the sun—credit growth far exceeds real economic growth, house prices far exceed residents' incomes—by Minsky's classification, this constitutes an early-warning signal of accumulating financial fragility. Ponzi finance denotes a financing position in which operating cash flows cannot cover even the interest due, requiring additional borrowing or asset sales to sustain payments. Minsky's (1992) Financial Instability Hypothesis asserts that, over periods of prolonged prosperity, economies tend to move from structures dominated by hedge finance toward structures in which speculative and Ponzi finance units carry a large weight—this is a tendency claim about structural evolution. The clock de-anchors automatically in tranquility—no external shock is needed, no one need act with malice.

But at least in this first de-anchoring, the clock remains anchored to a physical entity—the house is still there, people still live inside, the value of the location still has a causal relationship with commuting distance and school district quality. The first de-anchoring primarily changes financing and valuation mechanisms, and its ecological effects are transmitted mainly through rents and development decisions—what distinguishes it from the second de-anchoring is the primary transmission mechanism, not that it "leaves living space entirely unaffected."

§3 The Second De-Anchoring: The Attention Economy Reshapes the City's Ecological Functions

The first de-anchoring was at the financial level—capital transformed real estate from a physical entity into tradable readings, but the real estate itself remained physical. The second de-anchoring is at the cognitive level—the attention economy transforms the city from a physical information-processing space into content material for online platforms.

This de-anchoring is deeper—because what it affects is not the financial function of real estate (asset appreciation), but the ecological function of the city (processing information, maintaining social relationships, providing a space for existence in the time dimension).

Törnberg (2025) systematically describes this process, and Banks (2023), from the perspective of American cities, reached a similar diagnosis: the city is remaking itself to cater to the modern desire of the attention economy for "authentic urban life"—the city is becoming a set optimized for social media. Specifically, Törnberg's (2025) investigation found: under the economic pressures of attention, the biases and logics of social media have been integrated into the city's underlying political economy, and digital communicability has become an important objective of urban design and architecture. Investors see themselves as being primarily in the business not of food, drink, or cultural experiences—but in the business of attention.

Translated into the projection-side and defense-side language of MCAF. **Projection Side**: The unlocked stock (C) is no longer the physical resources of the city—it is the user data and attention patterns accumulated over decades by the internet. The media conductivity coefficient (Λ) is no longer the walkability of the streets—it is the recommendation algorithms of social media. Narrative load (ϕ) is no longer "is this neighborhood suitable to live in"—it is "does this space look good on social media." **Defense Side** degrades synchronously. Meta-audit capacity (A): a person's judgment of whether a neighborhood is good originally depended on walking around the street, smelling it, chatting with the neighbors—direct contact in the time dimension. Now it depends on check-in posts on the platform—readings in the clock dimension. Biological substrate (E_c): 10 bits/s (Zheng &

Meister, 2025) is reallocated from "observing the real environment on the street" to "scrolling through check-in photos on the phone." Legitimacy base ($\|L\|$): Performance legitimacy (L_{perf}) shifts from "does it feel good to live here" to "how high is the tag heat of this location."

Boy and Uitermark (2017) provided visual evidence of this impact: scrolling through Instagram, one sees "a seemingly interminable series of peak moments"—pictures taken and curated to convey that their posters are "happy, healthy and hip"; social media highlights aspects of the city that perform well on the platform while obscuring content that performs poorly. In the language of the Clock Room: all clocks show the same time—"this city is wonderful." Not because the city truly is wonderful—but because the parts that are not wonderful do not generate transmissible readings and disappear from the information environment. The effect of narrative load (ϕ) lies not only in controlling what people see but also in eliminating the contrast conditions necessary for auditing—when all visible readings point to "this city is wonderful," the person loses the frame of reference for judging deviation.

Naja, Valdez, and Antonini (2024) confirmed, through case-study analysis, a persistent feedback loop: the material form of the city is now in a continuous, mutually reshaping feedback loop with the digitally mediated practices driven by the attention economy. The clock points to the clock. The sun—the real use value of the city—vanishes from the mapping chain.

The Evidential Boundary of "Replacement." It should be noted that current evidence (Törnberg, 2025; Banks, 2023; Boy & Uitermark, 2017) demonstrates that the attention economy is **reshaping** urban design and economic logic but has not yet sufficiently demonstrated that this influence has constituted a **complete replacement** of the city's ecological functions. To demonstrate complete replacement, the following evidence would be needed: (1) historical downward trends in data on people's actual behavior in the city (walking time, frequency of face-to-face interaction); (2) a causal relationship (not merely correlation) between these declines and the intensity of platform use; (3) controlled variable analysis to rule out other factors (popularization of the car, suburbanization, increase in working hours, etc.). This evidence is not yet sufficient. Therefore, the "second deanchoring" should be understood as a process that is unfolding, not a completed state. The argumentative goal of this chapter is not to declare that "replacement has already been completed"—but to identify the mechanisms and direction of replacement, so that intervention can occur before replacement is complete.

§4 The Three-Step Manufacture of Pseudo-Small Fibers

§3 described how the attention economy is reshaping the city's ecological functions. But "reshaping" is only the surface—the fiber framework can reveal a deeper mechanism: the Discrete Leviathan does not replace the city's information-processing functions out of thin air, but manufactures a **pseudo-small fiber** with a very small perceived fiber but a very large real fiber—a cognitive simulacrum that is "more precise," "more real-time," "more personalized" than the city. It is precisely this simulacrum that leads large numbers of users to shift their attention and information acquisition from the time dimension to the clock dimension.

Step One: Capturing Attention—Manufacturing a Zero-Friction Information Environment. The physical space of the city possesses natural friction: it takes time to walk to the street, one needs to push the door to enter a shop, chatting with the shop owner requires presence. These frictions are the "cavity of the bowl"—the barriers that prevent a person's attention from being completely captured by the algorithm. The Discrete Leviathan substantially reduces these operational frictions through infinite scroll, autoplay, and one-click ordering. The

efficiency with which a user acquires information on the platform far exceeds the efficiency of acquiring information on the city streets—this manufactures a perceived fiber that "acquiring information online is faster and more accurate." But a pseudo-small fiber arises not from speed or display precision alone, but when users attribute more certainty to the readings than they warrant for the judgment at hand: the readings appear to have a very small fiber (precise, low latency), but the actual fiber is very large (information is filtered by commercial interests, the mapping function is unknown and unstable).

Step Two: Re-encoding Urban Space into Standardized Readings. The corner shop is a small-fiber clock—the mapping function is transparent, stable, and local. When the Discrete Leviathan re-encodes urban space into ratings, check-in posts, and short videos, it replaces a small-fiber mapping with a large-fiber mapping. The "4.8-star restaurant" seen by the user on the phone appears more "precise" than walking around the street and judging with one's own senses—this is the typical form of a pseudo-small fiber: the precision of the readings conceals the inflation of the fiber. The user does not know which population group these ratings come from, does not know whether the check-in posts are ranked by the algorithm, does not know whether "looks good" means "tastes good." The opacity of the mapping function is concealed by the precise appearance of the readings.

Step Three: Eliminating the Contrast Conditions Necessary for Auditing. When a sufficient number of urban spaces are presented on the platform interface in the form of standardized readings, the independent frame of reference available to users narrows accordingly. Old-style neighborhoods are marginalized by the algorithm for lacking "check-in value"—users cannot scroll to them, and thus do not consider them to exist. The hidden subspaces of the city (neighborhoods unsuitable for photography but suitable for living, communities without internet-famous shops but with close neighborly ties) are systematically suppressed in the platform interface; only the parts that conform to the algorithm's preferences are allowed to enter the visible subspace. This is not merely information filtering—it is the very conditions of audit being narrowed (the effect already stated in §3: when visible readings converge, the frame of reference for judging deviation disappears).

The effect of superimposing the three steps is: the city readings the user obtains on the platform appear more precise, more comprehensive, and more efficient than firsthand experience, but in reality, these readings are highly skewed, filtered by commercial interests, and reflect only a standardized projection of the city's visible subspace. The manufacture of the pseudo-small fiber is complete—the user feels unprecedented information abundance in the clock dimension while losing direct causal contact with the true state of the city (T) in the time dimension. ("Manufacture complete" here refers to the logical closure of the mechanism, with each step supported by independently observed instances; for the evidential boundary of replacement progress, see the end of §3.)

§5 The Discrete Leviathan: The Manifestation of the Four Criteria at the Urban Scale

The definition of the "Discrete Leviathan" is given in Chapter 25, §1—it is not a list of companies, nor a label of a certain capital market segment, but a network of key nodes defined by four discriminant conditions (serial control, substitution cost asymmetry, trans-sovereign reach, and capital structure reflexivity). This section does not redefine this object but merely identifies the concrete manifestation of the four criteria at the urban scale.

A reading convention must first be established: the formulations in this section and §11 that take "the Discrete Leviathan" as the grammatical subject (e.g., "eliminates friction through infinite scroll," "re-encodes urban space into standardized readings") refer to the effects structurally presented by the network of nodes hit by

the discriminant function—including the attention structures of algorithmic distribution, trans-sovereign data deposition, and the chain-multiplier pricing of the node narratives by capital markets—and do not refer to any unified, conscious actor. Competition, divergence, and different business models exist among these nodes.

Within the scope of this chapter, the four criteria are concretely manifested as three sets of observable consequences: (1) capital structure reflexivity and the capitalization output of digital unlocked stock (C_{dig}) at the capital market end—the aggregate market capitalization of the listed portion has surpassed the GDP magnitude of most nation-states; (2) a trans-sovereign reachable digital substrate—the same batch of infrastructure operates across multiple sovereign jurisdictions; (3) the structural replacement of the information-processing functions of the city's physical space by pseudo-small fibers (§4)—this is the urban-scale consequence on which the remaining sections of this chapter focus. The survey by Eisfeldt and Schubert (2025) provides adjacent academic evidence: corporate groups with higher AI talent concentration experienced a concurrent rise in systematic risk (Babina et al., 2023), a research direction discussed in connection with equity-market concentration and the relative dominance of the Magnificent Seven—but this group effect is a **concomitant phenomenon** hit by the discriminant function, not the discriminant condition itself.

First, capitalization output at the capital market end. Eisfeldt and Schubert (2025) report a related historical benchmark with a different scope: the "Artificial" portfolio, comprising firms in the highest quintile of labor-based generative-AI exposure, had a market capitalization of approximately $10.2 trillion on November 29, 2022. This portfolio was not constructed using this book's node-selection criteria. As of early October 2025, the several U.S. technology companies aggregated by the capital markets under the label "Magnificent Seven"—Nvidia, Microsoft, Apple, Alphabet, Amazon, Meta, and Tesla—had an aggregate market capitalization of $20.8 trillion, surpassing the $19.4 trillion GDP total of the EU-27 (Euronews, 2025). Nvidia alone had a market capitalization exceeding $4 trillion (Benaich et al., 2025), roughly equivalent to the GDP of Germany (Euronews, 2025).

A scope delimitation is needed here: the Mag 7 is a capital market label and does not fully coincide with the discriminant function of Chapter 23, §1. Nvidia (CUDA ecosystem), Microsoft (Azure and models), Alphabet (Search and Android), Apple (iOS gateway), Amazon (AWS), and Meta (social platform gateway) clearly strongly satisfy the two criteria of "serial control" and "substitution cost asymmetry"; Tesla, however, strongly satisfies mainly the "capital structure reflexivity" criterion (the narrative load of FSD/Robotaxi/Optimus is deeply coupled with capital expenditure expansion), and its strength on serial control (alternatives exist in both the automotive and autonomous driving supply chains) and substitution cost asymmetry (switching costs for durable consumer goods are not high) is markedly lower than the other six. This book, in using the figure of $20.8 trillion, treats it as a **reference for capital-market scale**—it includes nodes such as Tesla whose strength on some criteria is limited, and excludes unlisted nodes (OpenAI, Anthropic, xAI, ByteDance, SpaceX/Starlink, etc.) as well as nodes like TSMC and ASML that are hit by the discriminant but do not belong to the Mag 7; it is therefore not a formally established lower bound for the network selected by the four criteria.

Between the closing values on November 29, 2022 (the day before ChatGPT's release), and October 2, 2025, the Nasdaq-100 price index rose by approximately 116.4% (computed from the FRED NASDAQ100 daily closing series, Federal Reserve Bank of St. Louis, 2026), and the capitalization wave flying the "Magnificent Seven" banner brought the Mag 7's combined market value to $20.8 trillion (Euronews, 2025)—the real-time pricing in capital markets of the chain-multiplier product of the scale of the digital unlocked stock (C_{dig}), the media conductivity coefficient (Λ), and the narrative load (ϕ).

Supplementary Note: The Observable Counterpart at the Data Factor Layer under China's Institutional Path. The above capital-market-observable portion (the $20.8 trillion aggregate market capitalization of the Mag 7) is the marketized manifestation of the digital unlocked stock C_{dig} under the U.S. private capital path; under China's institutional path, the observable counterpart at the data factor layer is primarily supplemented by industry statistics and base construction—public materials from the National Data Administration show that the national data annotation industry exceeded RMB 8 billion in scale in 2024, and that the seven data-annotation bases had developed 524 datasets by the first half of 2025 (Gov.cn, 2025; Guizhou Fabu, 2025); the industry presents a coexistence of three organizational logics: "expert judgment / engineered production / platform-enabled empowerment" (Li Jingjing & Li Zihao, 2026). The two paths differ in capitalization mechanisms and in what is observable, but are both concrete manifestations of C_{dig} at the institutionally observable layer. This book only borrows the above industry statistics and organizational descriptions as observable-layer evidence and does not also borrow normative policy claims such as "sustainable infrastructure supporting national-level data factor production"—the latter belongs to the governance stance layer and is not conflated with the argumentative level of this section's structural diagnosis of the discriminant function.

Second, expansion rate. According to the plans summarized in the February 6, 2026 report, Amazon, Microsoft, Meta, and Google alone planned capital expenditures of approximately $650 billion in 2026—about 63% above the approximately $400 billion in 2025, which itself was about 67% above the approximately $240 billion in 2024 (Baab, 2026, computed from the listed amounts). On April 29, 2026, Microsoft guided to approximately $190 billion in capital expenditures for calendar 2026 (Microsoft, 2026). Differences in the treatment of leased assets and lease payments mean that companies' capex guidance is not necessarily comparable with cash expenditures on property and equipment. Stanford HAI (2025) reported the scale of capital flows: U.S. private AI investment reached $109.1 billion in 2024—nearly 12 times that of China and 24 times that of the UK; training compute doubles every 5 months, dataset size doubles every 8 months, and training power consumption doubles every year. The expansion is not limited to capital—it is physicalizing: multi-gigawatt data center clusters have moved from slideware to site plans (Benaich et al., 2025), and Microsoft signed a twenty-year power purchase agreement with Constellation, the operator of the Three Mile Island plant, paving the way for the reactor's restart to secure computing power supply (Eisfeldt & Schubert, 2025; Constellation, 2024). The Discrete Leviathan operates not only through digital systems: its computing demand is helping shape power development and reactor restarts through long-term contracts.

Third, information-processing capacity and attention shaping. The city's information-processing capacity comes from human-to-human interaction in physical space—the information flows generated by hundreds of thousands to tens of millions of people in streets, offices, and markets. The Discrete Leviathan's information-processing capacity comes from human-to-algorithm interaction in digital space—the data streams generated by billions of users' every click, swipe, and dwell. The two differ substantially in reach, recording granularity, and feedback speed—platforms operate on high-frequency interaction data from billions of users, far beyond what face-to-face interaction among hundreds of thousands to tens of millions of urban residents can produce and retain—though user counts alone do not establish a comparable throughput ratio. The replacement goes beyond throughput. González de la Torre, Pérez-Verdugo, and Barandiaran (2026), from the perspective of embodied cognitive science, argue that digital platforms do not merely passively process user attention, but actively shape the mode of attention itself through AI and big data analysis—"disaggregating habits away from the individual, into an AI managed collection of behavioural patterns." The result is a self-reinforcing loop: attention capture → data

extraction → platform tuning → recapture. When the operational speed of this loop far exceeds the information refresh rate that face-to-face interaction in the city's physical space can sustain, the replacement shifts from a quantitative advantage to a qualitative reorganization—the person no longer merely "does not need to walk into the city to acquire information"—the very organization of their attention is also being profoundly reshaped by the platform.

Törnberg (2025) provides the judgment for this process: "the imperative of data accumulation has produced an era in which everything—even our cities—is shaped by the pursuit of attention." This replacement is unfolding along the path of "the shift from the time dimension to the clock dimension" at the urban scale (for the evidential boundary of replacement progress, see the end of §3).

§6 Office Real Estate: Structural Divergence under Multi-Factor Superposition

The impact of the Discrete Leviathan on the city's physical space is not theoretical—but its measurement faces the dilemma of multi-factor attribution. The real estate downturn is the product of the superposition of multiple factors: the credit cycle, the interest rate environment, policy regulation, demographic structure, remote work technology, e-commerce substitution, and more. The task of this section is not to prove that remote work is the dominant factor—but to identify, among the multiple factors, which are cyclical (receding with financing conditions) and which are structural (altering demand over the longer term; long-lasting change does not necessarily imply irreversibility).

Facts. The U.S. office property vacancy rate is at a structural high in 2025: CBRE forecasts a full-year 2025 peak of approximately 19% (CBRE, 2025); Moody's Analytics measured the national office vacancy rate rising to a record high of 20.7% in Q2 2025 (Moody's Analytics, 2025)—its economists LaSalvia and Metcalfe predicted in mid-2024 that the vacancy rate would reach a peak of approximately 24% in 2026 (CFO Dive, 2024). San Francisco's office vacancy rate hit an all-time high of 36.9% in Q3 2024 and has since declined continuously with the demand expansion of the AI industry—falling to 32.8% in Q4 2025 and further to 29.2% in Q2 2026 (CBRE, 2026a, 2026b). The commercial mortgage-backed securities (CMBS) office loan delinquency rate rose to 11.01% at the end of 2024 (Trepp, 2024).

Multi-Factor Attribution. The study published by Gupta, Mittal, and Van Nieuwerburgh (2026) in the *American Economic Review* combines leasing data with a valuation model and estimates that the remote-work shock is associated with a decline of approximately 46% in the long-run value of New York City office buildings and approximately $557 billion in value destruction across the U.S. office market (AER original text: "$556.8 billion"). However, they also point out that it is necessary to distinguish between standard business cycle risk and the structural impact of remote work—rising interest rates significantly increased financing costs, and the downward phase of the economic cycle was superimposed upon the structural downward shift in demand. Surplus office space usually results from the combined effect of supply-side overbuilding and multi-factor contraction on the demand side; remote work is one of the factors, not the only one.

Evidence of Structural Change. San Francisco's peak vacancy rate of 36.9% can be explained by many factors—the massive tech layoffs of 2022–2023, high California taxes driving corporate relocation, and deteriorating public safety. But its diagnostic significance lies not in proving that remote work is the primary cause—but in revealing: even under conditions of economic recovery and falling interest rates, office demand in the tech industry—the industry that was earliest in completing the construction of remote collaboration infrastructure—

may not return to pre-pandemic scale, location, and patterns of use even where it recovers. As of March 2023, daily office utilisation in Asia-Pacific (the share of workers going to the office daily) stood at about 86%, versus 61% in EMEA and 48% in the Americas (Savills, 2023)—a measure of attendance behavior that should not be equated with property vacancy. This geographic divergence is consistent with an explanation involving differences in the adoption of remote collaboration, but the cited figures do not directly test that factor; Savills also emphasizes the role of office culture.

The Positioning of This Chapter. Cyclical factors (rising interest rates, economic recession, oversupply) explain "why the crisis erupted at this moment." Structural factors (remote collaboration replacing the information-exchange function of physical co-presence) explain "why, even if the cycle warms, the past may not return." This chapter focuses on the latter—not to negate the former, but to reveal the one dimension that standard analyses systematically underestimate. The "information-processing premium" embedded in office rents—the premise that people must be in the same physical space to exchange information—is being replaced by the Discrete Leviathan at near-zero marginal cost. When this premium is stripped away, the physical assets that carried the old function enter a revaluation that may prove long-term.

§7 Street-Level Shops: Pseudo-Small Fibers Eliminate Small-Fiber Clocks

If office vacancies reflect the reorganization of production space, the disappearance of street-level shops reveals a deeper form of regression—the loss of cognitive calibration function in living space. The mechanisms, evidence, and social consequences of the two are different.

Independent Shops Are Small-Fiber Clocks. The small shop on the corner is also a kind of clock—it has business hours, prices, product structure—but its mapping function (π) is transparent, stable, and local. The person knows what this shop sells, why it sells at this price, who the shop owner is. The fiber is small. Brand chain stores, by contrast, are standardized large-fiber clocks—the mapping function is formulated by headquarters; pricing, product selection, and store design are all controlled by remote algorithms; the fiber is large. The small-fiber advantage here arises from durable local relationships and repeated opportunities for verification; it is not guaranteed by independent ownership alone—Boy and Uitermark (2017), cited in this chapter, also observe that some independent boutique establishments receive more tags on the platform than chain stores. This section concerns shops whose community functions are strong but poorly represented by platform visibility. But the pseudo-small fiber diagnosis of §4 reveals a dynamic process more important than static classification: the Discrete Leviathan does not simply "replace" small-fiber clocks with large-fiber clocks—it compresses the opportunities of some small-fiber clocks to enter users' information environments through the three-step manufacture of pseudo-small fibers.

The Positive Feedback from Marginalization to Elimination. The following describes a possible feedback pathway where shops depend heavily on platforms for customer acquisition and other channels cannot adequately compensate. A community café that values neighborly relations finds it difficult to fully present its community value through ratings alone; a traditional craft business with little online promotion and limited customer-generated or third-party information may receive little exposure in users' recommendation feeds—the visibility of these shops on the platform interface is persistently reduced and may approach zero in extreme cases: the former cannot display its community value within the rating system, while the latter's platform visibility is thereby limited. When visibility is compressed to near zero, the shops do not exist in the visible

subspace: users do not know about them—not because they have closed, but because the platform does not show them. Customer traffic is diverted elsewhere by the algorithm; revenue falls; this may contribute to eventual business exit. After closure, foot traffic on the street decreases, and the information-exchange function of the remaining shops (daily neighborly conversations, serendipitous contacts) degrades accordingly—less information exchange means more shops are marginalized. This is the fiber diagnosis of the Doom Loop: the pseudo-small fiber creates a self-fulfilling prophecy at the reading end and, through a positive feedback loop, converts the prediction at the reading end into reality at the real end.

Wolton et al. (2025), using postcode-level outlet and ward data from the UK for 2011–2019, found an association, supported by a causal identification strategy (panel plus instrumental variable), between the decline of independent consumption-type shops and the rise in the vote share of UKIP (the UK Independence Party); the paper's additional tests show that the effect is driven by long-established outlets and by function categories that can serve a social role—pubs, personal services, food and drink, multi-item retail, and, to a lesser extent, household and related goods outlets (e.g., book stores)—the independent consumer outlets the paper uses to approximate "third places." This differential effect points not to the absence of "commercial function"—brand chains can substitute for commercial function—but to the disappearance of the small-fiber mapping function itself. The social capital production function borne by independent shops—daily conversations between people and shop owners, chance encounters among neighbors within the shop, information exchange unmediated by any algorithm—constituted the physical infrastructure of meta-audit capacity (A) at the community level. What the pseudo-small fiber eliminates is precisely this infrastructure.

Fetzer et al. (2025), using high-street vacancy data, independently found this association: there is a significant positive correlation between high street vacancy rates and UKIP support. They emphasize that high street vacancy is a "particularly visible form of local decline"—unlike office vacancies, the vacancy of street-level shops is directly exposed to public view. Every pulled-down roller shutter is physical evidence of the degradation of the city's ecological function.

The Magnitude of the Doom Loop. *The Guardian* (2026) reported the acceleration of the positive feedback: approximately 37 shops closed per day in the UK in 2024, with nearly 13,000 permanently closed over the full year. Fetzer describes the phenomenon as "You almost have a non-linear tipping point" beyond which high streets cascade into oblivion. The closure statistics show the scale of exit; judgments about critical points and self-reinforcing processes belong to the researchers and to this chapter's interpretation of the mechanism of decline—this chapter offers a mechanism hypothesis for this process through the pseudo-small fiber framework and does not treat the closure figures themselves as a verification of that mechanism.

Boundaries of Attribution. What must be made clear is: these studies support the social and political significance of retail change—Wolton et al. provide causal-identification support for the link between the decline of independent outlets and rising UKIP vote share, along with evidence for the social-capital mechanism; Fetzer et al. find a robust association between high-street vacancy and UKIP support. Whether platform ranking contributes to shop exits is not what these studies directly identify. The demise of independent shops is the result of the superposition of multiple factors: the impact of e-commerce, rent increases, the economic cycle, policy changes, and more. This chapter accordingly proposes a mechanism interpretation: where customer acquisition depends heavily on platforms and community functions cannot fully translate into platform visibility, reduced visibility, business exits, and declining street-level interaction may reinforce one another. This interpretation does not require platforms to be the sole or dominant cause.

The OECD (2026) treats local retail as the social infrastructure of the community (community cohesion, local identity, local vitality), while also discussing the conditions under which digital tools can support physical retail—the report supports protecting offline social functions, but does not support the conclusion that digitalization will inevitably eliminate them. Frago (2026) builds on and develops the concept of "retail-less cities" proposed by Carreras and Frago (2022)—not cities without retail in the literal sense, but a historically situated urban condition in which retail loses its structural role in organizing the city, with logistical logics increasingly assuming the intermediary function.

§8 Ghostification: The Back-Staging of the City

The most extreme form of the second de-anchoring is not the vacancy of shops, but shops that still exist physically yet have already been "ghostified" in their cognitive and social functions. Ghostification and shop exits may both be connected to platforms' reorganization of customer flows and fulfillment, but they need not operate through the same visibility mechanism. Section 7 concerns reduced online visibility; this section concerns a diminished offline public interface, even where platform visibility and order volumes remain high. A pseudo-small fiber arises when users overestimate how adequately online readings represent actual operations or public functions: the shops are not vacant, but their functions as ecological clocks may be severely impaired. In other words, ghostification is not an alternative to disappearance, but another form of disappearance—an invisible decline that does not physically manifest as "vacancy."

"Ghost shops" are a spatial continuum that exists parasitically on the Discrete Leviathan: **dark kitchens**—the umbrella category delineated by Huang et al. (2025), comprising two forms: **ghost kitchens**—non-customer-facing, multi-tenant shared commercial kitchens, mostly located in industrial or warehouse-type properties, where each food tenant has its own staff and equipment but typically shares storage and ancillary facilities (Huang et al., 2025); and **virtual brands**—brands operating out of existing street-level physical kitchens and solely through delivery platforms, accounting for 89.1% of dark kitchen businesses in England (Huang et al., 2025); in addition, there are **dark grocery stores**—micro-fulfillment centers not open to the public and dedicated to fulfilling online orders; and **delivery-only stalls**—the Chinese variant documented by Jiang Xiaomei and Ji Deqiang (2024): bypassing organized operational forms such as cloud kitchens and shared kitchens, operated by individuals making full use of the city's interstitial and underground spaces—some renting small storefronts of a few square meters along streets and alleys, open only with a single pickup window facing the street, where delivery riders pick up and go; others are centrally managed in uniformly branded "food courts." Food Court A on the B1 level of Songjiazhuang Subway Station in Beijing's Fengtai District—located under Beijing's largest affordable housing community and the intersection of three subway lines (Line 5, Line 10, and the Yizhuang Line)—is a typical spatial carrier of this form (Jiang Xiaomei & Ji Deqiang, 2024).

In the Western context, Huang et al. (2025), using a sample of 113,370 online food outlets in England (covering the three major platforms of Uber Eats, JustEat, and Deliveroo), identified 7,054 dark kitchens (housing 15,809 businesses) and 470 dark grocery stores (in 362 locations)—businesses operating from dark kitchens accounted for 15.9% of all non-grocery online food outlets. The study reveals the unequal distribution of ghost formats: England's "dark retail" is disproportionately concentrated in areas of greater socioeconomic deprivation—at the LSOA level, the mean number of online food outlets operating from dark kitchens in the most deprived quintile was approximately 3.60 times that in the least deprived quintile (unadjusted model); the ratio was 2.34 after adjustment for population density.

In the Chinese context, Jiang Xiaomei and Ji Deqiang (2024) conceptualize this phenomenon as "platformized migration"—delivery-only merchants are simultaneously driven by two kinds of migration: a "digital migration" from the "dine-in + delivery" model to a pure delivery economy, and a "physical migration" to specific urban spaces (upper-floor crevices and underground B1 levels)—the result being that "in the process of 'stallization,' the city folds itself, forming 'vast worlds within the folds' " (author's translation). The convergence of the Chinese and Western cases points to the same spatial inequality—platforms utilize low-cost, low-visibility community spaces as logistics nodes, and in some neighborhoods this may reinforce a spatial orientation of substituting logistics fulfillment for public interfaces of encounter.

More gravely, these ghost shops are becoming regulatory blind spots—Huang et al. (2025) suggest that adding virtual brands onto existing kitchens may expose a planning gap based on the approval of new physical outlets: in policy areas designed to restrict the expansion of traditional fast food (such as takeaway management zones around schools), the association between adopting zones and the number of dark kitchens is not statistically significant after controlling for population density and area deprivation, and whether evasion is widespread remains to be tested; in China, the "food courts" on the B1 levels of subway stations and the few-square-meter window storefronts along streets and alleys are equally difficult to incorporate into existing food safety supervision frameworks.

The multi-layer order forwarding and forged credentials also provoked concentrated enforcement by regulators against the platforms. Two related but distinct phenomena must be distinguished here: one is the conversion of physical premises toward delivery fulfillment and away from public-facing interfaces (dark kitchens and dark grocery stores mostly belong to this type, and the existing kitchens on which virtual brands rely may even continue to serve dine-in customers); the other is the appearance on platforms of shops lacking authentic operators or valid credentials—the enforcement case below belongs to the latter, revealing problems of traceability and platform vetting rather than the disappearance of some public storefront. In July 2025, the Haidian District Market Supervision Administration of Beijing, starting from a cake complaint, discovered that the 378 chain stores under the brand name "Tianyan Qingshu" (甜颜情书, "Sweet Love Letter") had no physical offline presence whatsoever, all their food business licenses were forged, and the same forged license number had been reused up to 10 times across different districts (Xinhua Net, 2026). Following the trail led to Chongqing's "Zhuan Dan Bao" and Anhui's "Xun Meng," two order-forwarding hubs that were granted open API interfaces by seven major e-commerce platforms, signed cooperation agreements with them, and placed orders on a "bidding auction." The special task force uncovered a total of over 67,000 "ghost takeaway" shops and over 3.6 million non-compliant orders (Xinhua Net, 2026). On April 17, 2026, the State Administration for Market Regulation imposed fines and confiscation of illegal gains totaling RMB 3.597 billion on seven e-commerce platforms—Pinduoduo, Meituan, JD.com, Ele.me (Taobao Flash Purchase), Douyin, Taobao, and Tmall—and combined fines of RMB 19.6874 million on the legal representatives and food safety directors of the seven platforms—the largest penalty imposed since the implementation of the *Food Safety Law* (Xinhua Net, 2026).

The significance of this enforcement action lies in this: law enforcement "successfully penetrated the platform data barrier, achieving the precise extraction, categorized sorting, and effective solidification of key electronic data on order routing, qualification filing, and order-forwarding transactions" (Xinhua Net, 2026; author's translation)—what this chapter calls the item-by-item settlement of the dark nodes: the dark nodes within the three-tier architecture of "platform → order-forwarding platform → ghost shop," originally uncountable to regulators, were re-placed into a countable category. This regulatory response cannot immediately rebuild the street

ecology, but it shows that the mapping opacity discussed in §4 can be partially corrected at the regulatory level: the traceability between platform records and actual operators is being restored, making dark fibers countable again, clearable item by item.

"Ghostification" reflects the minimalist demand the Discrete Leviathan places on offline space—no longer requiring traditional display fronts, street-facing windows, and customer entrances, demanding only that the space fulfill the functions of storage, sorting, and instant delivery. Physically, the shop still exists (it is not vacant), but for premises converted from public-facing storefronts to pure fulfillment spaces, their function as ecological clocks—capturing attention, exchanging information, sustaining community publicness through a physical storefront—may be substantially weakened. When human faces and street vitality are replaced by algorithmic rankings and delivery routes, the shop still performs production and fulfillment functions, but no longer provides a public interface of encounter for residents. The public values carried by the corner shop—social interaction, community engagement—are discarded as "redundancy"; behind digital convenience lies the withering of real interpersonal networks.

§9 Three Irreversibilities: Three Consequences of the Pseudo-Small Fiber

After the superposition of the two de-anchorings, the three irreversibilities—information loss, dynamic corrosion, ontological dimensionality reduction—present a unique morphology in urban space. But the pseudo-small fiber diagnosis of §4 reveals a crucial supplement: these three irreversibilities are not parallel consequences of the city "being replaced"—they are three facets of the manufacture of the pseudo-small fiber and its positive feedback loop.

First Layer (Information Loss) = The Direct Product of the Pseudo-Small Fiber. Step Three of §4—eliminating the contrast conditions necessary for auditing—is precisely the concrete mechanism of information loss. The city's traditional information processing provided direct causal contact with the true state (T). After the Discrete Leviathan replaces this channel, what the person obtains is entirely mapped readings, and the input of the mapping has been pre-filtered—the recommendation algorithm determines what the person sees and what they do not see. Not only is the mapping function unknown, but the input of the mapping has also been filtered by commercial interests.

Second Layer (Dynamic Corrosion) = The Positive Feedback of the Pseudo-Small Fiber. The Doom Loop of §7 is the unfolding of dynamic corrosion at the street scale. Two kinds of adaptive bias advance simultaneously: strategic gaming—developers deliberately design "internet-famous check-in spots," not because this design makes people live better, but because social media exposure can be converted into sales volume; selective adaptation—old-style neighborhoods maladapted to social media aesthetics are marginalized in the information ecology—not because anyone is suppressing them, but because the algorithm does not recommend them. The urban spaces that survive are not those better suited for living—but those better suited for photography.

Third Layer (Ontological Dimensionality Reduction) = The Projection Mechanism of the Pseudo-Small Fiber. The state space of the city's true state includes the physical quality of buildings, the density of community social relations, the accessibility of public spaces, the sense of environmental safety, cultural diversity, the frequency of intergenerational interaction, and the walking experience of the street. The pseudo-small fiber projects this state space onto the dimensions the Discrete Leviathan can process: photographability, ratability, communicability.

$$\dim(\text{The True State of the City}) \gg \dim(\text{The Observation Space of the Discrete Leviathan})$$

The dimensions discarded by the projection happen to be the core functional dimensions of the city as an ecological clock—dimensions such as neighborly trust, belonging, and serendipitous connections generated in aimless wandering may be omitted or compressed into inadequate proxies in interfaces organized around exposure and engagement metrics. The city's "non-being"—the *Tao Te Ching* uses the cavities of three objects as analogies: the emptiness of the hub (the unoccupied portion of the biological substrate E_c), the emptiness of the vessel (the work space of meta-audit capacity A), the emptiness of the room (the channel for causal contact with the real)—is nearly invisible in the mapping of the Discrete Leviathan. Not concealed—not on the priority list of dimensions in the platform's observable space.

The convergence of the three layers points to a judgment: the irreversibility manufactured by the pseudo-small fiber lies not in any single layer standing alone, but in the positive feedback among the three layers—information loss makes it impossible to recognize dynamic corrosion, dynamic corrosion makes urban space adapt to the preferences of the pseudo-small fiber, and ontological dimensionality reduction makes the discarded dimensions liable to long-term loss. Once this spiral is set in motion, simple "digital governance" or "platform regulation" can hardly reverse it—because the discarded dimensions are not concealed (recoverable through information disclosure) but are being physically hollowed out (shops close, neighbors move away, streets become desolate). Once the hollowing-out reaches a critical point, any correction at the information layer has already lost the object of reconstruction.

§10 The Real Estate Crisis as the Settlement of De-Anchoring: Global Isomorphism and National Variants

The database covering 66 countries and eight centuries by Reinhart and Rogoff (2009) indicates that excessive debt accumulation during booms often poses greater systemic risk than it appears, and that housing price cycles have long played a central role in financial crises. Brunnermeier, Rother, and Schnabel (2020) provided the magnitude: during real estate busts, the 95th and 99th percentiles of the estimated increase in systemic risk (ΔCoVaR) in dependence of balance-sheet characteristics were equivalent to about 55% and nearly 70% of the sample median of ΔCoVaR, respectively; the same paper emphasizes that this risk increment depends heavily on banks' balance-sheet characteristics—the transmission of systemic risk from real estate bubbles is not uniform. In the depth of embedding between the financial clock and physical rhythms, real estate is a salient asset class—it embeds the financial clock deeply into the rhythms of physical construction, population migration, and daily dwelling, and is therefore prone to structural settlement when rhythms are mismatched.

In the context of the Clock Room thought experiment, the structural pattern of every real estate crisis is identical: the financial clock first de-anchors from time—readings cease to have causal contact with the true state; then the clock turns to self-reference and self-reproducing "baroque-ification," with readings referring only to other readings, internal circulation replacing external calibration; the fiber then expands to the full set, the uncertainty of the mapping function expands to cover all possible states, and readings lose information content; until an external shock or an endogenous breakdown in financing exposes the underestimated risks, and settlement arrives.

Japan 1990: The Clock of Trend Extrapolation. The mapping function was: "Tokyo's land is limited + the

economy is growing rapidly → land prices can only rise, not fall." This is the macro version of the inference fallacy—correct pacing (the economy was indeed growing) was mistaken for an accurate starting point (land prices reflected true value). What the clock smoothed over was the expectation that two layers of real rhythm—demographic structure and the technology adoption of physical industry—would keep changing: Comin (2008) documented the post-bubble decay of these rhythms—the total factor productivity growth rate fell from an annual average of 2.4% for 1983–1991 (still high growth during the bubble period) to 0.2% for 1991–2000; the annual GNP growth rate per working-age person collapsed from 4.6% for 1960–1990 to 0.7% during the "lost decade"; R&D expenditure ceased to expand; the adoption speed of new technologies—chiefly computers and the internet—was overtaken by South Korea. The deceleration occurred after the bubble burst—the slope formed during the growth period was therefore persistently disappointed; this is the consequence of a rhythmic-layer mismatch rather than its precondition, and one of the conditions for the chronicity of the settlement.

Japan is therefore a chronic settlement: no concentrated $M(\tau^*)$ jump occurred—shadow debt was not, on a single date, cashed out all at once into visible losses, but was dilutively digested over thirty years through the low-intensity seepage of zombie enterprises, continuous land devaluation, and annual bank write-downs; meta-audit capacity (A) perpetually lacked a sufficiently sharp trigger signal to rebuild the mapping between readings and the true state. Rogoff and Yang (2026) calculate that the value of Japan's housing stock at its 1990/1991 peak was approximately $20 trillion, remaining less than four times the contemporaneous stock market capitalization (approximately $5 trillion)—this scale gave the correction broad balance-sheet consequences, and its duration also depended on financial resolution, demographic change, and investment adjustment.

U.S. 2008: The Clock of the Probability Model. Common one-factor Gaussian copula specifications describe default dependence through a limited set of parameters and shared risk factors—they do not assume unconditional independence across borrowers. The concern here is not that such models contain no temporal information, but that simplified dependence assumptions may be treated as stable across periods and market conditions. An even deeper layer of de-anchoring occurred in the incentive structure of the securitization chain: Keys, Mukherjee, Seru, and Vig (2010) demonstrated that securitization separated loan origination from the ultimate bearing of default risk, weakening some originators' incentives to screen borrowers carefully—the delinquency rate on low-documentation loans just above the FICO 620 hard threshold was roughly 20% higher than the sample's mean default rate of 5% (about a 1 percentage point increase) within two years of origination—despite these loans' theoretically slightly better credit profile—and delinquencies in the highly securitized subprime market of 2005–2007 rose by approximately 50%. Loan underwriting—that embodied process, the face-to-face information exchange between loan officer and borrower—was compressed by the securitization chain into a standardized score; the reading (FICO score) became detached from the true state (the borrower's repayment capacity). This is a textbook case of dimensionality reduction. Acute settlement—the Lehman collapse as the $M(\tau^*)$ jump, forcibly converting the shadow debt accumulated over years within the room into cracks on the wall.

China: Adjustments to High Turnover and Administrative Calibration since 2020. The high-turnover model of Chinese real estate developers can be traced back to 2010—what brokers called the "inaugural year" of high performance and high turnover—Vanke being the earliest to implement the "5986" model (land acquisition to construction start in 5 months, sales in 9 months, 80% sold in the first month, with 60% of products required to be residential), and Country Garden subsequently pushing it to the extreme with the "456" rhythm (land acquisition to project launch in 4 months, capital recovery in 5 months, capital re-circulation in 6 months)

(Jia Guoqiang, 2018)—high turnover compresses the rhythm of project launch and capital recovery, not the physical construction cycle (the launch construction period and the building construction period are two different things). Zhang Weiping and Cao Tingqiu (2022) academically summarized this phenomenon as a "high debt, high leverage, high turnover" industry-wide operational model, not the isolated strategy of a single developer; both Country Garden and Evergrande were representative developers deeply cultivating third- and fourth-tier cities under this model (Zhang Genxue, 2021). Local governments increasingly relied on land sales to obtain fiscal revenue and further used land as collateral to obtain financial resources, forming a "land finance—land-based finance" cycle, at the cost of high housing prices and mounting local government debt (Zhao Fuyang, Chen Binkai, & Liu Shouying, 2021; Rogoff & Yang, 2022). Chang, Wang, and Xiong (2026), based on a quantitative study of 173 cities from 2015–2022, further provided the scale of this cycle: in 2019, land sales accounted for 38.1% of local government fiscal revenue (with significant cross-city variation), and cities that were more reliant on land sales and land-backed debt during the pandemic saw more significant price increases and price-volume divergence.

Since 2020, the emphasis of China's real estate policies has shifted, but deleveraging, risk resolution, and market stabilization have overlapped rather than formed mutually exclusive phases. Financing constraints from 2020—the "Three Red Lines" policy of August 2020—tightened the financing space through which highly leveraged developers sustained high turnover via debt expansion; by 2022, financing support and measures to ensure housing delivery were already operating alongside risk resolution. This is precisely the institutional-layer meta-audit mechanism argued in Chapter 3, §3.2—macro-level meta-audit capacity (A_{macro}) does not attempt to catch up within the clock dimension, but modifies the projection-side variables through the direct exercise of power—instantiated in China's real estate scenario, and belonging to the same structural-intervention path as the asset management new rules of Chapter 22, §3.3 (the simultaneous intervention in the media conductivity coefficient, narrative load, and meta-audit). After 2024, the emphasis on market stabilization strengthened further—the policy focus shifted to city-specific measures, destocking, and revitalizing existing stock. In the survey's detailed breakdown, housing accounted for 59.1% of the total assets of the urban households surveyed in 2019, while housing loans accounted for 75.9% of total household liabilities (People's Bank of China Survey and Statistics Department Urban Residents' Household Assets and Liabilities Survey Task Force, 2020)—when real estate is simultaneously the household's store of wealth (financial clock) and living space (ecological clock), the shockwave of this settlement penetrates both dimensions.

§11 The State and the Discrete Leviathan: An Asymmetric Nesting Relationship

In the narrative of the two de-anchorings, one structural question of vital importance has not yet been directly confronted: what exactly is the relationship between the state—Hobbes's Leviathan—and the Discrete Leviathan?

The state possesses sovereignty—the monopoly on violence and the power to make law. The platform possesses algorithms—the monopoly on data and the distribution of attention. The nature of power of the two is completely different; they are not on the same plane. To juxtapose the state and the platform as "two Leviathans" would create a false symmetry—as if the two were on the same tier of power. In reality, the platform's servers are built on the state's territory, its contracts are protected by the state's laws, and its tax obligations are stipulated by the state. The $20.8 trillion market-value reference for the Mag 7 listed in §5 likewise depends on the legal and physical infrastructures provided and maintained by states, resting on the state's tolerance.

This relationship is an asymmetric nesting, not a symmetric competition. The platform operates parasitically within the state's physical and legal infrastructure; it does not seek to "replace" the state (Hobbesian Leviathan competition) but seeks only to bypass regulation on key functions. When the platform's expansion threatens the state's core interests—the tax base, data sovereignty, ideological control—the state directly smashes the clock: China's 2020–2021 platform antitrust crackdown, the EU's General Data Protection Regulation (GDPR, 2018), the U.S. antitrust hearings and lawsuits (2020–2025). On November 3, 2020, the Shanghai Stock Exchange decided to suspend the STAR Market listing of Ant Group—the state can change the rules of the game at any time.

But the state's regulation is systematically lagged by the structural mismatch of time scales. Here a three-tier clock-mismatch chain emerges:

- **The Sovereign Clock** (legislative cycle: yearly scale): The legislative process requires hearings, deliberation, voting, and implementation; the time from problem discovery to policy effectiveness is typically measured in years.
- **The Financial Evaluation Clock** (quarterly reporting and some financing reviews as representative examples): This section uses quarterly evaluation as one illustration of market-level rhythm; it should not be equated with the 16–20 year medium-term financial cycle identified by Drehmann et al. (2012) as discussed in §2. Losing sight of the medium-term cycle and failing to keep pace with quarterly rhythms are different phenomena.
- **The Algorithmic Clock** (iteration cycle: daily to hourly scale): Platform algorithm updates occur on a daily or even hourly basis; a single recommendation response is counted in milliseconds, but the observation period of an A/B test runs to days or even weeks. The rhythm of algorithm deployment and effect observation is far faster than the financial clock, and far faster than the legislative cycle.

$$\tau_{\text{Sovereign}} \gg \tau_{\text{Financial}} \gg \tau_{\text{Algorithmic}}$$

(The clocks in this equation are scenario-specific expressions used in this chapter: the sovereign clock corresponds to the legislative dimension of the governance clock (τ_g) of Chapter 3, §3.2; the financial clock and the algorithmic clock describe the rhythms of the projection side at the market and algorithmic scales, and may be understood by analogy to the scenario projection of the technology-capital clock (τ_c).)

This is not a problem of "inadequate regulation"—it is a structural decoupling between two types of clocks. The "sovereign clock" cannot outrun the "algorithmic clock" not because legislators are incompetent, but because the time scale of rulemaking, evidence gathering, and enforcement is systematically slower than the rhythm of algorithm deployment and effect observation. The Digital Services Act took a little over three years from its proposal in December 2020 to its full applicability in February 2024—this process shows that rule formation takes time, but no estimate of how many times the algorithms have iterated can be derived from it. Operational speed does not mean regulation is necessarily futile—regulation need not chase every parameter update, but can instead constrain objectives, accountability, and auditability.

Financialization transformed real estate from a physical entity into tradable readings (the first de-anchoring). The attention economy transformed the city from a living space into material for virtual traffic flows (the second de-anchoring). The two de-anchorings overlap in time, coexist in space, and mutually reinforce each other

in mechanism—but they both occur under the state's tolerance or failures of oversight. The degradation of urban space's vitality is a process that the "asymmetric collision" between state and platform may accelerate—an absent or insufficient compensation loop and regulatory lag are among the conditions that permitted its acceleration; whether they contribute more than rents, planning, demographics, labor organization, or the business cycle is not something this chapter's evidence can adjudicate. The state could have, much earlier, constrained the platform's replacement of the city's physical space through antitrust, improvement of the compensation loop, and data protection—but the structural sluggishness of the sovereign clock left it at risk of acting too slowly, too late.

§12 The Final Line of Defense of the Time Dimension

The reason real estate is perpetually at the center of financial crises (Reinhart & Rogoff, 2009; Brunnermeier et al., 2020) is not because it is the largest asset class—though it is. It is because it is the piece of physical existence hardest to move fully into the clock dimension—property rights, debt, and prices can be recorded digitally, but construction, maintenance, dwelling, and encounter still require physical space. Stocks, bonds, derivatives can be fully digitized—they were abstract clocks from the beginning. Attention, social relations, knowledge acquisition can be moved online—their digitized versions are close to functional substitutes. But the body cannot be moved into the screen. A person must live somewhere. The body must occupy physical space.

The physical irreplaceability of real estate is the final anchor of the time dimension within the economic system—what Mehrling (2015) described as "an open-ended future that is substantial uncertain [sic] (not just risky) and a present choice set that is substantially constrained by survival constraints of various kinds." The body must be sheltered. This constraint cannot be financialized, cannot be algorithmized, cannot be discretized.

The real estate crisis is therefore not merely a financial event—it is the collision at the final boundary between the clock dimension and the time dimension. When the financial clock attempts to encode the full value of real estate into tradable readings, it strikes an impenetrable wall: inside the house live people; people have bodies; bodies exist in time; time is not a clock. These attempts at compression—Japan's land price myth, America's CDO^2, China's high turnover—ended with the drastic expansion of the fiber, i.e., crisis; the three cases show that such mismatches can have severe consequences, but not that every temporal compression under financialization must end in crisis.

The rise of the Discrete Leviathan adds a new variable: it not only attempts to encode the financial value of real estate into readings (the first de-anchoring) but also attempts to replace the ecological function of the city itself (the second de-anchoring). The settlement of the first de-anchoring manifests as price collapse—financial crisis. The settlement of the second de-anchoring may not manifest as price collapse—but as the structural degradation of urban functions. Regional and sectoral divergence in office demand (based on the U.S. national and San Francisco data used in this chapter), shifts of some consumer spaces toward delivery fulfillment, and the association between retail exits and local decline—these provide the empirical entry points for this chapter's account of the second de-anchoring's settlement forms; their global generality and counterpart forms in other space types (e.g., shopping malls) require separate evidence (for the evidential boundary of replacement progress, see the end of §3). The political correlates of independent-shop decline and high-street decay can already be observed and measured; the erosion of social capital is one important mechanism among

them that has received research support. The degradation of the city's ecological functions is not an abstract theoretical prediction—it has already been recorded in ballots.

"Serving the real economy through finance" acquires its deepest meaning in this context: **protecting the city's "non-being."** Above, the *Tao Te Ching* was already invoked to argue the three levels of "non-being" (the emptiness of the hub, the vessel, and the room).

In urban space, these three forms of "non-being" have physical counterparts. The bench in the park—has no commercial function, produces no GDP, need not aim at producing social media content, but is the breathing space of the biological substrate (E_c)—when a person sits there, they can temporarily step back from continuously responding to platform readings, existing in time. The small shop on the corner—not an internet-famous check-in spot, not recommended by the algorithm, but it is the work scene of meta-audit capacity (A)—the daily conversation between the person and the shop owner is an information exchange that bypasses platform ranking and commercial encoding—a mapping function that is largely known and repeatedly verifiable. The plaza with no defined function—produces no efficiency, and its public value cannot be exhausted by its degree of monetizability, but is the channel for causal contact with the real—there, the person feels the rhythm of the city, not the rhythm of the algorithm.

Protecting these spaces is not nostalgia. It is not the romanticization of inefficiency. It is protecting all the conditions on which the defense side depends for its operation. The design direction of the projection side—faster media conductivity coefficient (Λ), fuller information flows, more seamless interaction—each of these eliminates the city's "non-being." Faster conduction fills the axle hole. More seamless interaction fills the cavity of the bowl. More comprehensive information coverage seals the window.

The pseudo-small fiber identifies a defense-side rationale for protecting "non-being." Park benches, plazas without a prescribed function, street corners not organized around check-ins—urban spaces that produce no readings, carry no ratings, and receive no algorithmic recommendation—are irreplaceable because they are where the user's perceived fiber is not compressed by platform readings. Under the definition of Chapter 4, §2.4 (The Pseudo-Small Fiber), a pseudo-small fiber is a state in which the true fiber is large while the perceived fiber is small; the three steps described in §4 (The Three-Step Manufacture of Pseudo-Small Fibers)—lowering operational friction, encoding urban space into standardized readings, and narrowing the contrast conditions needed for auditing—each presuppose that the user is inside the platform's information environment. Daydreaming on a park bench takes place outside that environment, where readings play no part in shaping perception; strolling through a plaza and neighborly trust can all be recorded, evaluated, and shared, but such readings do not adequately represent their cognitive and public value—so long as people do not invest these readings with more certainty than they actually support, a pseudo-small fiber does not arise. The significance of protecting these spaces lies not in any natural immunity to algorithms, but in preserving forms of public use that do not require platform visibility, ratings, or immediate monetization as conditions of access. As long as these conditions hold, people retain the opportunity for independent observation, interaction, and verification beyond platform readings.

This is structurally similar to the human biological substrate ($E_c \approx 10$ bits/s, a research estimate of the information throughput of human behavior, whose neural substrate remains unexplained; Zheng & Meister, 2025)—both are constraints that existing technology can hardly accelerate significantly, both are materializations of the time dimension. The biological substrate delimits the information throughput of the individual; the city's

"non-being" delimits the cognitive calibration space of the community. When the former is filled up by the platform interface, and the latter is displaced by commercial use, platform distribution, and attention allocation, the conditions for the defense side's operation are drastically weakened—these spaces cannot be adequately represented, but the spaces that carry them can nevertheless be squeezed out.

"[W]hat is there provides benefit and how what is not there provides functionality" (Laozi, *Tao Te Ching*, Wang Bi recension, ch. 11; Lynn trans., 1999). The city's "being"—skyscrapers, commercial complexes, internet-famous landmarks—is "benefit." The city's "non-being"—ruins, vacant lots, uncommunicable everyday life—is "functionality." The former lends itself to financialization and to being priced by the attention economy; the latter, even where it can be developed or disseminated, has public functions that cannot be adequately represented by such pricing. But the latter is precisely the source of the city's function as an ecological clock.

The direction of repair—if one exists—lies fundamentally in: **protect "non-being."** The only possible safety mechanism depends on what von Neumann (1955) called "an intelligent exercise of day-to-day judgment" (von Neumann, 1955; for the quotation and translation analysis of this term, see Chapter 1). This chapter applies the principle to the city: this intelligent exercise does not take place on the screen—it takes place on the street. It does not take place in the algorithm—it takes place in the body. It does not take place in the clock dimension—it takes place in the time dimension.

The city's streets are the last window. To protect them is not nostalgia—it is because, in a room without a window, no number of clocks can tell you what time it is.

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